

CU Boulder - Data Science Capstone Project Description



99P Labs / Honda Research Institute US, Inc.

Company Address: 1255 Kinnear Rd, Columbus, OH 43212

Industry Description and Company Products: <https://www.99plabs.com>

Primary Contact(s)/Mentor(s) Names and Emails:

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Planned Meeting Schedule with Students

Meeting Venue (Google Meet or similar)

Meeting Occurrence Rate (weekly)

Attention Measure: We will be available to help the students brainstorm and work through problems, offering guidance and support when needed. However, we encourage the team to take agency over the project and make it their own. Our goal is to foster an environment where students can innovate, make independent decisions, and drive the project based on their ideas and insights. We will have weekly meetings to discuss progress and challenges, and we'll maintain open communication through Slack for quick questions and continuous collaboration.

Requested Team Size: 3-5

Specific Required/Expected Deliverables:

- A functional MVP (Minimum Viable Product) of a system that accepts data samples or schemas and generates a report..
- Comprehensive documentation, including system architecture, user manuals, and technical guides.
- A final presentation showcasing the MVP's features, capabilities, and potential impact.
- A detailed project report outlining the development process, methodologies used, challenges encountered, and solutions implemented.
- A blog post summarizing the project and its outcomes, to be published on the 99P Labs Medium account.

Required Intellectual Property: none

Project Introduction: Many organizations possess datasets but struggle to identify how to extract meaningful insights or initiate data-driven projects. This capstone project centers on "Data Science as a Service," focusing on data mining. The objective is to develop a system that leverages Large Language Models (LLMs) to analyze a sample of data and generate a report outlining potential data science projects that could be undertaken with the data. By inputting a data sample or schema, users will receive guidance on possible analyses, insights, and applications, effectively unlocking the value of their datasets.

The project's focus areas include natural language processing, machine learning, data analysis, and user interface design. Stakeholders include 99P Labs, the student team, and potential end-users who will benefit from actionable data science project ideas generated from their own data.

List of Project Functionalities (Functional Elements – what will be created, built, examined, discovered, predicted, etc.)

- **Data Input Module:** Ability to accept various data formats (CSV, JSON, SQL schemas) as input.
- **LLM Integration:** Utilize LLMs to interpret the data structure and content.
- **Prompt Engineering and Model Fine-Tuning:**
 - Develop and optimize prompts to effectively guide the LLMs in generating accurate and relevant analyses.
 - Explore the possibility of fine-tuning LLMs on domain-specific data to enhance performance and tailor outputs to specific use cases.
- **Automated Reporting:** Generate detailed reports outlining:
 - Potential statistical analyses.
 - Data visualization recommendations.
 - Machine learning models applicable to the data.
 - Insights on data quality and preprocessing needs.
- **Use Case Idea Generation:** Provide users with potential use cases and applications for their data based on its content and structure, helping them understand how the data can be leveraged in real-world scenarios.
- **User Interface:** Develop a user-friendly interface for data input and report viewing.
- **Scalability Considerations:** Design the system to handle different data sizes and complexities.
- **Extensibility:** Ensure the system can be extended with additional functionalities in the future.

List Dataset Needs and Availability:

Students must create or gather data. Public Datasets: Use of open-source datasets from platforms like Kaggle or UCI Machine Learning Repository.

Project Notes, Requirements, Expectations

Project Background and Objective: 99P Labs aims to lower the barriers to data science by creating tools that make data analysis accessible to non-experts. The objective of this capstone project is to develop an MVP of a system that employs LLMs to provide insightful analyses and actionable recommendations based on user-provided data samples or schemas. This tool will help users understand the potential locked within their data without requiring advanced data science expertise.

Project Requirements:

- **Functional MVP Development:** Create a working prototype that meets the outlined functionalities.
- **User-Centric Design:** Focus on usability to ensure the tool is intuitive for users with limited technical backgrounds.
- **Timely Delivery:** Complete the project within the allocated 15-week timeframe.
- **Testing and Validation:** Implement thorough testing to ensure reliability and accuracy of the generated reports.
- **Blog Post Creation:** Develop a comprehensive blog post detailing the project, to be published on the 99P Labs Medium account.

Why students should choose this project:

- **Innovative Experience:** Work with state-of-the-art AI technologies and contribute to making data science more accessible.
- **Skill Development:** Enhance skills in machine learning, NLP, software development, and user experience design.
- **Impactful Outcome:** Create a tool that has the potential to assist a wide range of users across various industries.
- **Mentorship:** Gain valuable insights and guidance from experienced professionals in the AI and data science fields.
- **Publication Opportunity:** Have your work showcased on the 99P Labs Medium account, gaining visibility in the tech community.
- **Collaborative Environment:** Engage in regular meetings and continuous communication via Slack to foster teamwork and professional growth.

Deliverables:

- **Functional MVP:** A fully operational prototype of the data analysis and reporting system.
- **Technical Documentation:** Detailed documentation covering system architecture, codebase, and user guides.
- **Final Presentation:** A comprehensive presentation demonstrating the system's features, benefits, and potential applications.
- **Project Report:** An in-depth report outlining the project's objectives, methodologies, results, and reflections.

- **Blog Post:** A well-crafted blog post summarizing the project and its outcomes, suitable for publication on the 99P Labs Medium account.

Skill Requirements:

- **Programming:** Proficiency in Python.
- **Machine Learning:** Understanding of machine learning concepts and frameworks (e.g., TensorFlow, PyTorch).
- **LLMs:** Experience with LLMs.
- **Data Analysis:** Skills in data preprocessing, statistical analysis, and data visualization.
- **APIs and Integration:** Knowledge of integrating APIs, especially for AI models.
- **Version Control:** Experience with Git or other version control systems.
- **Soft Skills:** Strong communication, teamwork, and problem-solving abilities.

Images: N/A

Other Questions to Consider and Note:

- 1) What is the nature of the project?
 - a. This project is development-focused with clear objectives to build a functional MVP. It combines software engineering with applied AI and offers opportunities for innovation and creative problem-solving within defined goals.
- 2) Do students need to be US Citizens to participate in the project?
 - a. No
- 3) What backgrounds or prerequisites would you prefer that students have? (This might include Python, R, Linear Algebra, writing skills, curiosity, etc.)
 - a. Curiosity and desire to experiment
- 4) How many hours per week do you expect from each student on your team?
 - a. 10
- 5) Please include here any links or data or websites that might help students to better understand the overall nature of the focus of the project.
 - a. **Kaggle Datasets:** <https://www.kaggle.com/datasets>
 - b. **UCI Machine Learning Repository:** <https://archive.ics.uci.edu/ml/index.php>
 - c. **Ollama:** <https://ollama.com/>
 - d. **Julius AI:** <https://julius.ai/>
- 6) In one paragraph, please describe the expectations you have of your student team with respect to the project, deliverables, meetings, engagement, etc.
 - a. We expect the student team to engage proactively with mentors and team members, providing regular updates and communicating openly about challenges and progress. The team should deliver high-quality work with attention to detail and best practices, be open to feedback and willing to iterate on solutions, manage time effectively to meet project milestones and deadlines, and utilize Slack for day-to-day communication and collaboration.

- 7) Please include anything else here that will help me, you, and students in creating appropriate and successful teams.